		SET-I	Total Pages: 4

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
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Discipline: INTERDISCIPLINARY

Course Code & Name: 221RGE100 : RESEARCH METHODOLOGY AND IPR

Max. Marks: 60		Duration: 2.5 Ho

PART A

		Answer all questions. Each question carries 5 marks	Ma
	1	With neat diagram, illustrate the different steps in research process.	(5
	2	Explain the different methods of information gathering.	(5
	3	Describe the role of hypothesis in experiment.	(5
	4	Enlist the guidelines for preparation of good presentation slides.	(5
	5	What are Intellectual Property Rights? How are they protected?	(5
	6	Depict pictorially the different stages from filing to grant of patent.	(5

PART B

Read the given article and write a report that addresses the following issues

7 **Human Assisting Robot**

Various works are carried out on Human - following robots, for their immense potential to carry out mundane tasks like load carrying and monitoring of target individual through interaction. The recent advancements in vision and sensor technologies have helped in creating more user-friendly robots that are able to coexist with humans by leveraging the sensors for human detection, human movement estimation. But most of these sensors are suitable only for Line of Sight Objects. In case of loss of sight of the target, most of them fail to reacquire their target. This work proposes a novel method to develop a human following robot using Bluetooth, GPS and Inertial Measurement Unit (IMU) on smartphones which can work under high interference environment and can reacquire the target if the target position is lost.

Recent developments in the robotics world have developed robots and are user friendly, intelligent, and most importantly affordable. With these benefits of robotics, it is no wonder that they have found to be used in every field. The benefits of robots have increased their flexibility with being capable of performing a variety of tasks and applications. Robots also allow for increased production and profit margin because they can complete tasks faster. Robots have the ability to work around the clock since they do not require vacations, sick days, or breaks. Robots are preferred over men doing the job specifically in hazardous environments, handling toxic substances, working with heavy loads and wherever repetitive tasks are involved. This work on human assisting robot uses Bluetooth to have communication between the robot and the tag held by the user, here after called target. In this work mobile is used as a tag. In the first literature discussing the applications of human assisting robots, generally used in hospitals, restaurants and airports, etc to carry luggage. However most of these robots fail to re-acquire their target, if line of sight is lost is discussed by second literature. This problem can be eliminated using GPS and GPS has the advantage of tracking the robot even its position data is lost. To enable the tracking system on robot, GPS is used along with the GY273 compass which helps in giving appropriate commands to the motor drivers. The primary goal of our work was to design and fabricate a robot that not only tracks the target but also moves towards it with the help of GPS and compass. The third literature discussing the disadvantage of using the GPS is that it does not work indoor. In this work all the processing is carried out by the Arduino UNO microcontroller and L298N motor drivers are used to drive the motors.

The circuit consist of Arduino UNO microcontroller, HMC5883L compass, NEO6m GPS, HC06 Bluetooth module, L298N motor driver, 12 V DC motors and batteries. The three challenges that are faced while assembling above components and making human assisting robot are,

(i) power consumption for driving the motor and other electronic systems
(ii) control system used to control the robot
(iii) tracking the robot.

To drive the motor 12V battery is used another 9V battery is used to power other electronic modules. For controlling functions Arduino UNO microcontroller which is based on ATmega328P is used. This acts as the brain of entire system. It receives the information from different modules such as compass, GPS etc, and checks information and identifies the location of target. Then it sends the instruction to motor driver for making movements. Then continued tracking is done using the data of GPS Neo 6m and magnetometer GY273. The position is determined by the GPS Neo 6m. Magnetometer GY273 is used to measure the direction of movement of target called heading.

In this work there is no governing body, i.e it does not require any human intervention for command or direction. This means all the work is done by the modules interacting among themselves as per the algorithm. Hence we can say that the Autonomous connection is established. Here Arduino is the brain of the entire system. It collects information from GPS, compass and Blynk platform via Bluetooth and give commands to the motor driver for controlling the robot. Everything starts with the connection with the Bluetooth. As the Bluetooth is connected to the smart phone it means there is a connection established between the target and the robot. For control and navigation GPS Neo 6m is used along with magnetometer GY273. The Arduino gets the position of the robot and its direction of movement from these modules. For robot movement, DC motors are used. The H bridge motor driver is used to control the speed and the direction of the DC motor. PWM signal is used to control the speed and input pins are used to control the direction. Whenever there is a change in the direction by the human the Arduino compares the position with and check the new direction with the help of magnetometer. Now it gives information to the motor driver through PWM pulse for changing the direction and follow the master.

- Firstly, by using compass the direction is known where it's pointing (p1). So, in relation to the north pole it is called heading represented as a1.
- Secondly, position of android device can be estimated (p2). So, we can find the distance between the two points by using the common formula as: -x1)2 -(y2-y1)2
- Angle between the point p2 and the pole is known as bearing. It can be represented as a2.
- The turn angle can be estimated as : t=a2-a1 or t=bearing-heading
- Turn angle is calculated in order to move towards a particular point or direction.

As expected the robot functions autonomously, without the intervention of the target / human. The algorithm written into the Arduino helps the robot to follow target by acquiring signals from sensors. Our robot was able to track the human (tag) independent of the direction (Left, Right, and straight) in which he was moving.

The paper has presented a Human-following robot which is being used for their immense potential to carry out mundane tasks like carrying load and for monitoring of an individual through interaction. However the present prototype is tested only with its movement on different direction and tracking the human who held the tag. In this work mobile is used as a tag. This system works efficiently in outdoor; however it is inefficient in indoors due to poor

		GPS signal. This work can be further improved by (i) adding obstacle detection sensors, (ii) scaling up the project for carrying weight and (iii) design wheels for moving in terrain surfaces. By adding the above features we can develop this work into product for public / commercial use.	
	a	What is the main research problem addressed?	(4
	b	Identify the type of research.	(4
	c	Discuss the short comings in the literature review if any?	(4
	d	Discuss the significance of the study.	(6
	e	Discuss appropriateness of the methodology used for the study.	(6
	f	Summarize the important results and contributions by the authors.	(6